Gait Analysis of Children with Cerebral Palsy *

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Executive Summary

Dr. Conaway is interested in demonstrating a significant connection between the total strength of youth with cerebral palsy (CP) and their walking speed. Establishing this relationship provides the necessary justification to expand the current study into a funded clinical trial. In Section 1, we detail the specific motivating questions regarding the relationship between strength, walking speed, and children's demographics. In Section 2, we describe the structure of the dataset and the imputation methods used to address missingness. In Section 3, we conduct exploratory data analyses of the children's walking speed, strength, and demographic distribution, specifically examining the ordinal nature of GMFCS levels. In Section 4, we perform a hierarchical linear regression analysis to isolate the association between strength and walking speed after controlling for covariates, while also testing for complex interaction effects. In Section 5, we shift the focus to a binary outcome of walking speed that is "adequate for community dwelling," and utilize logistic regression to model how strength influences the probability of achieving this functional benchmark. Finally, Section 6 summarizes the key takeaways and provides specific design recommendations for a future clinical trial. Supplementary figures and model diagnostics can be found in the Appendix.

(1) Introduction

The goal of this report is to provide evidence regarding the association between `walking speed` and `strength`, and to better understand how these variables relate within the context of childrens' conditions. While a simple correlation would show the basic association, we must use the available demographic covariates to explain as much variation in `walking speed` as possible before isolating the relationship with `strength`. Furthermore, we investigate how `strength` interacts with these covariates to determine if complex relationships exist that researchers should be aware of when designing a clinical trial. Finally, we analyze the concept of a `walking speed` that is "adequate for community dwelling," providing necessary context regarding the functional benchmarks children with cerebral palsy (CP) aim to achieve.

(2) Data

The dataset consists of gait lab measurements from boys and girls with CP visiting Shriners Hospitals for Children in Lexington, Kentucky. Dr. Conaway considers this sample representative of the larger CP population. The dependent variable for this analysis is `walking speed` (recorded as "Walking_Speed"), and the primary variable of interest is `strength` (recorded as "TotStrengthSum").

Overview

Including records with missing data, there are 352 subjects in the dataset between the ages of seven and nineteen. Although there are five `GMFCS` levels, only the three least severe levels are represented in this study. For `CP Type`, we combined the levels *Hemiplegia (right)* and *Hemiplegia (left)* into a single level

*Formatting based on svmille guide (http://github.com/svmiller).

(*Hemiplegia*) since the involved side is not relevant to the subjects' CP severity and therefore the outcome of `walking speed`.[1]

The variable `strength` is a composite of strength measurements taken at individual joints, measuring the force exerted during movements such as ankle or knee extensions. These measurements were recorded before children walked a distance of approximately 50 feet three times, with breaks in between. In most cases, the average was used to determine the single value for `walking speed`. However, refusals were omitted; Dr. Conaway is confident these instances reflected a lack of willingness rather than an inability to walk. Additionally, the median was used if one of the three measurements was a clear outlier.

Missingness

There are 31 records with a total of 35 missing observations in the dataset. To test if this missing data is Missing Completely at Random (MCAR) we can perform Little's MCAR Test.

```
## # A tibble: 1 x 4
##   statistic    df p.value missing.patterns
##       <dbl> <dbl>   <dbl>            <int>
## 1      51.8    31  0.0110                5
```

With a p-value of 0.0110 (< 0.05), we reject the null hypothesis that the missing values are purely accidental with no underlying pattern. As a result, we cannot just drop these observations, and should favor an imputation method appropriate to Missing at Random (MAR) data (Dr. Conaway's comments indicate the pattern is not Missing Not at Random).

Imputation

Predictive mean matching works by fitting a regression model for the variable with missing observations against the other variables. Then, for each missing entry, it identifies the complete records with the closest predicted values. A random selection is made from the closest actual observed values to fill the missing entries, which preserves the variance better than imputing an average.

All subsequent analyses, which were done with the imputed data, were also run without imputation (where we just dropped any records with missing data), and no contradictions were found between analysis with and without imputation.

(3) Exploratory Data Analysis

We first present counts, summary statistics, and distributions, then address potential multicollinearity.

Summary

Table 1 shows the distribution of children in the dataset across levels of `GMFCS` and `CP Type`:

Table 1: GMFCS Level vs CP Type

	Diplegia	Hemiplegia	Sum
1	61	72	133

[1] Kulak and Sobaniec (2004) paper.

	Diplegia	Hemiplegia	Sum
2	120	24	144
3	75	0	75
Sum	256	96	352

A possible concern, which we will try to address later by checking if certain levels can be merged, is the zero count of children with `GMFCS` level 3 Hemiplegia CP. The absence of data for this subgroup limits the analysis, as any potential relationship specific to this combination of levels cannot be estimated (`GMFCS` and `CP Type` interaction).

The mean of the different quantitative variables in the data is given in Table 2:

Table 2: Means of Quantitative Variables

Variable	Mean
Walking Speed	92
Strength	713
Age	13
Height	148
Weight	45
Walking Age	28

These means indicate that the variable units are comparable in magnitude, so no coefficients will be confusingly scaled during the subsequent analysis. Note that I refer to the units for `strength` as "units" because there was not clear confirmation from the client that the units are Newtons.

The histogram of `walking speed` (Figure 3) displays a distribution that is approximately normal. The values range from roughly 10 cm/sec to 160 cm/sec, with a slight left skew indicating a small tail of children with a significantly lower `walking speed`.

Next, the scatterplot of `walking speed` against `strength` (Figure 4) reveals a positive linear association. However, the relationship is characterized by substantial variability, as indicated by the wide spread of data points around the regression line. While higher `strength` generally corresponds to a faster `walking speed`, the variability suggests that other factors likely contribute to the variation in `walking speed`.

Lastly, Figure 5 reveals that while there is substantial overlap between the distributions of `GMFCS` levels 1 and 2, level 3 is much lower. Notably, there is no overlap between the IQR of Level 3 and the IQRs of the other two levels. This trend confirms the ordinal nature of the variable, as `walking speed` decreases across quantiles with increasing `GMFCS` severity.

Multicollinearity

To choose the covariates that explain any variation strength does not, we first created a correlation matrix to find which covariates are strongly linearly related. Note that `BMI` is not included in this matrix since it is directly constructed from the variables `height` and `weight`, and I was more interested in the different contributions both those variables made.

Table 3: Correlation Matrix of Covariates

	Age	Height	Weight	Strength	Walking Age
Age	1	0.8	0.57	0.24	0
Height	0.8	1	0.72	0.42	-0.11
Weight	0.57	0.72	1	0.36	-0.09
Strength	0.24	0.42	0.36	1	-0.31
Walking Age	0	-0.11	-0.09	-0.31	1

The results in Table 3 show the strongest correlation is with age and height (0.8), with the next largest correlations between height and weight (0.72) and age and weight (0.57). Therefore, one of the covariates age, height, and weight should be considered to be dropped during model fitting because they explain so much of the same information. Also, there is a correlation between strength and walking age (0.41), albeit a weak one; this is promising for our motivating question about if an association between strength and walking age exists.

(4) Regression Analysis

To address the main question in this report, we now perform hierarchical regression analysis to show if strength explains a statistically significant amount of variance in walking speed after accounting for all other covariates[2].

Hierarchical Linear Regression

The first model (Model 1) includes the demographic information, starting with all of the potential covariates:

$$\begin{aligned}\text{walking_speed} = \alpha + \beta_1(\text{age}) + \beta_2(\text{height}) + \beta_3(\text{weight}) + \beta_4(\text{walking_age}) + \beta_5(\text{gender}_{\text{Male}}) + \\ \beta_6(\text{gmfcs}_2) + \beta_7(\text{gmfcs}_3) + \beta_8(\text{cptype}_{\text{Hemiplegia}}) + \epsilon\end{aligned} \tag{1}$$

The highest VIF values for the covariates in Model 1 are from height (4.48), age (2.96), and weight (2.18). All of these VIF values are less than 5, so there is not enough evidence of multicollinearity to justify removing any of these demographic covariates.

In the next step (Model 2), we add the variable of interest, strength:

$$\begin{aligned}\text{walking_speed} = \alpha + \beta_1(\text{age}) + \beta_2(\text{height}) + \beta_3(\text{weight}) + \beta_4(\text{walking_age}) + \beta_5(\text{gender}_{\text{Male}}) + \\ \beta_6(\text{gmfcs}_2) + \beta_7(\text{gmfcs}_3) + \beta_8(\text{cptype}_{\text{Hemiplegia}}) + \beta_9(\text{strength}) + \epsilon\end{aligned} \tag{2}$$

Our interest is whether Model 2 explains the walking speed better than Model 1. Here, we have a 0.015 increase in R^2, and using the function *anova()* gives a corresponding F-statistic of 10.73 and p-value of 0.001 to an increased R^2. Therefore, we reject the null hypothesis that both models explain the same amount of variation in walking speed and conclude that strength has a significant association with walking speed. Table 4 summarizes these results and the significance of specific predictors:

[2] Kim (2016) article.

Table 4: Hierarchical Regression Analysis (*p<.05, **p<.01, ***p<.001)

Covariates	Model 1	Model 2
Intercept	66.18***	67.90***
Age (years)	-2.81***	-2.63***
Height (cm)	0.55***	0.46**
Weight (kg)	-0.06	-0.11
Walking Age (years)	-0.14	-0.11
Gender (male)	-2.28	-2.79
GMFCS Level 2	-8.93***	-6.95*
GMFCS Level 3	-41.79***	-38.09***
CP Type (hemiplegia)	3.10	3.51
Strength		0.01**
R-Squared	0.50	0.52
R-Squared Change	0.503	0.015
F Change		10.73

Model 2 Assumptions

Next, the assumptions for Model 2 need to be checked using the Figure 4 graphs:

- Linearity (Residuals vs Fitted): The red LOWESS line is nearly perfectly horizontal and the points are scattered randomly above and below the line, so the relationship between the predictors and walking speed is linear.
- Normality of Residuals (Normal QQ Plot): The points follow the line very closely for the vast majority of the data. There is a tiny bit of deviation at the very top right, but this is not extreme so we can conclude that the residuals follow a normal distribution.
- Homoscedasticity (Scale-Location): The red LOWESS line is flat and the spread of the points in the left cluster looks roughly similar to the spread in the right cluster, so the variance of the errors is constant.
- Influential Outliers (Residuals vs Leverage): Observation 310 has a high leverage but does not exceed Cook's distance and is therefore kept.

Standardized Coefficents

While the coefficient for `strength` in Model 2 has the smallest magnitude at first glance, this is due to the differing units. The standardized model can be calculated to compare the effect sizes of the different covariates:

$$\widehat{\text{walking_speed}} = 0 - 0.27(\text{age}) + 0.25(\text{height}) - 0.06(\text{weight}) - 0.06(\text{walking_age}) - 0.05(\text{gender}_{\text{Male}}) - 0.13(\text{gmfcs}_2) - 0.58(\text{gmfcs}_3) + 0.06(\text{cptype}_{\text{Hemiplegia}}) + 0.15(\text{strength}) \tag{3}$$

Previously, the interpretation of $\beta_9 = 0.01$ was that for a one unit increase in `strength`, a child with cerebral palsy's `walking speed` increases by 0.01 cm/sec. This directly answers the motivating question by showing there is a significant positive relationship between `strength` and `walking speed`.

The standardized $\beta_9^* = 0.15$ is interpreted as for every one standard deviation increase in strength, walking speed increases by 0.15 standard deviations. This removes the units and shows the relative importance of strength on walking speed compared to the demographic covariates. The value 0.15 is the fourth largest standardized coefficient behind the GMFCS level 3 indicator, age, and height.

This demonstrates the importance of strength, especially considering the other top predictors are either purely biological traits tied to stride length (age and height) or an indicator of the most severe GMFCS level that also cannot be changed. Unlike these factors, strength is modifiable, which clearly motivates clinical interventions targeting strength.

Pooling Factors

Before creating interaction models, we consider the possibility of merging two of the three GMFCS levels. This would simplify the model and resolve the zero cell count for Hemiplegia at GMFCS level 3. Table 5 shows the Model 2 indicator coefficients for GMFCS:

Table 5: Model Coefficients for GMFCS Levels

GMFCS Level	Coefficient (Standard Error)
1	0.00 (ref)
2	-6.95 (2.72)
3	-38.09 (3.61)

Since the Model 2 coefficient estimators are asymptotically normally distributed, we use Wald tests to determine if GMFCS levels can be pooled. We test if level 2 differs from the baseline ($H_0 : \beta_6 = 0$) and if level 3 differs from level 2 ($H_0 : \beta_6 = \beta_7$). The significant p-values (0.011 and $< 2 \times 10^{-16}$) indicate the levels are distinct; therefore, we reject both null hypotheses that the levels can be merged and keep all three.

Interaction Models

Next, we fit an interaction including terms for strength by CP Type and strength by GMFCS to determine if the association between strength and walking speed depends on these covariates. As expected, the CP Type interaction in Model 3 was not significant, similar to its main effect in Model 2. While insignificant demographic covariates are generally retained because our goal is inference rather than prediction, we removed this specific interaction term because it is no longer just background information and there is no evidence of a relationship.

After refitting the model with only the strength and GMFCS interaction (Model 4), a significant interaction exists between strength and GMFCS level 3. The p-value of 0.01 falls below the Bonferroni-corrected cutoff of $\frac{0.05}{3} \approx 0.017$, used to control Type I error for the three planned tests of interactions. The positive coefficient for this term indicates that the effect of strength on walking speed is significantly stronger for patients with GMFCS level 3 compared to those with less severe conditions.

Here is the final OLS model (Model 4):

$$\begin{aligned}\text{walking_speed} = \alpha &+ \beta_1(\text{age}) + \beta_2(\text{height}) + \beta_3(\text{weight}) + \beta_4(\text{walking_age}) + \\ &\beta_5(\text{gender}_{\text{Male}}) + \beta_6(\text{gmfcs}_2) + \beta_7(\text{gmfcs}_3) + \beta_8(\text{cptype}_{\text{Hemiplegia}}) + \beta_9(\text{strength}) + \\ &\beta_{10}(\text{gmfcs}_2 \times \text{strength}) + \beta_{11}(\text{gmfcs}_3 \times \text{strength}) + \epsilon\end{aligned} \tag{4}$$

Interesting Intercept Model

With the final model identified, we make one last modification to increase interpretability: creating an interesting intercept by centering the continuous covariates. This forces the intercept to represent the average child in the dataset, rather than a theoretical child with zero `age`, `height`, `strength`, etc.

Model 5 now shows a predicted `walking speed` of 104.57 cm/sec for a child of average `age`, `height`, `weight`, `walking age`, and `strength` This baseline is important for the next section, where the focus shifts to determining the attainability of a `walking speed` of 80 cm/sec. Furthermore, this ensures the main effects of the interaction terms reflect group differences at the mean level of `strength`, rather than at a `strength` of zero.

(5) Binary Outcome Analysis

Shifting focus to practical outcomes, a `walking speed` of 80 cm/sec is considered "adequate for community dwelling". To see if `strength` affects this new outcome, we now fit a logistic regression model starting from the covariates that have already been deemed appropriate in the previous regression section.

Logistic Regression

Model 6 is a logistic regression with the outcome 1 for an adequate `walking speed` greater than or equal to 80 cm/sec:

$$\begin{aligned}\log\left[\frac{P(\text{adequate_speed} = 1)}{1 - P(\text{adequate_speed} = 1)}\right] = \alpha &+ \beta_1(\text{age}) + \beta_2(\text{height}) + \beta_3(\text{weight}) + \\ &\beta_4(\text{walking_age}) + \beta_5(\text{gender}_{\text{Male}}) + \beta_6(\text{gmfcs}_2) + \beta_7(\text{gmfcs}_3) + \\ &\beta_8(\text{cptype}_{\text{Hemiplegia}}) + \beta_9(\text{strength})\end{aligned} \tag{5}$$

The coefficient for `strength` in this model is statistically significant ($p = 0.0005$), showing an association between adequate `walking speed` and `strength`. The interpretation is that for a one unit increase in `strength`, the odds of achieving an adequate `walking speed` increase by a factor of $\exp(0.002) \approx 1.002$ (or about 0.2%).

As a sanity check, we plot (Figure 1) the probability of adequate `walking speed` by `strength`. There appear to be substantially more children predicted to have an adequate `walking speed`, which we will discuss in a future section, but generally the fitted curve appears correct with its S-shape. Further investigation shows that only 3.7% of subjects had a predicted probability of adequate `walking speed` between 0.4 and 0.6. Therefore, there is no concern that the specific cutoff of 80 is creating many edge cases where shifting the requirement would drastically change the classification results. For these reasons, we keep this logistic regression approach going forward.

Check for Interaction

Once again, we check for a significant interaction. This time we start by fitting an interaction with just the term for `strength` by `GMFCS` (Model 7), and neither of the interaction terms are significant here (p-values of 0.61 and 0.20). This means that the association between adequate `walking speed` and `strength` does not depend statistically on the type and severity of CP. We continue the rest of the analysis with Model 6 that has no interactions.

Model 6 Assumptions

Logistic regression does not require the same assumptions as OLS, but we still need to assess:

- No Multicollinearity: The assumption is satisfied as the largest VIF is 4.44, which is below the threshold of 5.
- No Influential Outliers (Residuals vs Leverage): While Observation 275 exhibits high leverage, it does not exceed the Cook's distance threshold and is therefore kept.
- Linearity of the Logit: The Box-Tidwell Test was used. None of the interaction terms between continuous covariates and their logarithms were statistically significant (all $p > 0.05$), indicating that the linearity assumption was satisfied.

Validating Model Accuracy

A confusion matrix is created to check the accuracy of the model:

Table 6: Predicted vs Actual Adequate Walking Speed

	Actual: No	Actual: Yes	Sum
Predicted: No (<80)	64	14	78
Predicted: Yes (>=80)	33	241	274
Sum	97	255	352

This model achieved an overall prediction accuracy of 86.65%. It correctly identified adequate walkers 94.51% of the time (sensitivity), but only correctly identified inadequate walkers 65.98% of the time (specificity). This disparity occurs because the dataset is imbalanced, containing nearly three times as many adequate walkers as inadequate ones. Consequently, the logistic regression prioritizes overall accuracy, biasing predictions toward the majority class.

This is not a significant concern, as we can use the pROC package to identify a cutoff that better balances sensitivity and specificity. By adjusting the threshold to 0.67 from 0.5, the overall accuracy shifts slightly to 85.80%, while sensitivity becomes 90.59% and specificity improves to 73.20%. Although the primary goal here is inference rather than prediction, this adjustment demonstrates how a future clinical model could be tuned to handle similar population imbalances effectively.

Strength Across GMFCS Levels

During EDA, we observed that the average `walking speed` in the dataset is 92, a finding consistent with the logistic regression results. Given this, it is important to investigate patterns among the underrepresented children falling below the adequate `walking speed` cutoff to identify those who might benefit from

targeted attention. The following dose-response curve shows how the predicted probability of adequate walking speed increases with strength, this time controlling for other covariates in the model, unlike in Figure 2.

Figure 2 shows how the probability of adequate walking speed changes with strength for a child of average age, height, etc. The pattern in this adjusted model reveals that, while increased strength improves the probability for children at GMFCS levels 1 and 2, it is the GMFCS level 3 children who actually cross the 50% probability threshold for the average child (at approximately 1000 units of strength). Although the slopes of these curves are not statistically different, as indicated by the non-significant interaction (note that the previous regression did have a significant interaction), the positions of the curves relative to 80 cm/sec are practically important. This finding provides the strongest evidence yet for a clinical trial, especially to test if strength training can help GMFCS level 3 children achieve the practical goal of adequate walking speed.

(6) Conclusions

I have divided the conclusion into a recap of the results from the analysis and a second section that presents recommendations for designing a clinical trial based on these findings.

Results

The analysis led to three primary findings regarding the relationship between strength and walking speed in children with CP.

1. Association Between Strength and Walking Speed: There is a statistically significant positive association between strength and walking speed, even after adjusting for other covariates. While demographic covariates such as age and height account for a large portion of the variance in walking speed (as expected in growing children), the hierarchical regression demonstrated that adding strength to the model significantly improved explanatory power ($p = 0.001$). The analysis of standardized coefficients revealed that strength is the fourth most influential predictor of walking speed ($\beta^* = 0.15$), ranking behind only GMFCS, age, and height. This suggests that strength is a critical, modifiable factor in determining walking speed, distinct from the natural increases associated with getting older and taller.

2. The Role of CP Severity and Type: The strength of the association between strength and walking speed depends on the severity of the condition (GMFCS level) but not the CP Type (Diplegia vs Hemiplegia).

- CP Type: We found no evidence that the relationship between strength and walking speed differs between children with Diplegia and Hemiplegia CP Type.
- GMFCS: There is a significant interaction between strength and GMFCS level 3 ($p < 0.05$). The positive interaction coefficient indicates that for children with level 3 CP, increases in strength are associated with greater gains in walking speed compared to children with level 1 or 2 CP.

3. Adequate Walking Speed for Community Dwelling: A child's strength is a significant predictor of achieving the "adequate for community dwelling" benchmark of an 80 cm/sec walking speed.

The logistic regression model indicates that for every one unit increase in strength, the odds of a child achieving an adequate walking speed increase by approximately 0.2%. Unlike the linear model, the logistic analysis did not find a statistically significant interaction between strength and GMFCS level

regarding the odds of success. However, the probability curves shows that an average child at GMFCS levels 1 and 2 have high probabilities of adequate walking speed regardless of strength, whereas an average child at GMFCS level 3 require significant strength gains to cross the 50% probability threshold for achieving adequate walking speed.

Clinical Trial Recommendations

Assuming a similar cross-sectional study is going to be designed based on this current study, I propose two modifications to improve the robustness of the results.

First, the study should better account for fatigue and varying levels of effort. Since children are currently instructed to walk at a "comfortable speed," the measurements likely reflect different levels of effort. For example, a stronger child might walk at a moderate pace that easily meets the "adequate" threshold, while another child might barely cross that threshold using maximum effort. To address this, the study could incorporate varying walking distances to assess gait sustainability. While the average speed could still be the primary metric, measuring the speed falloff over longer distances would create a valuable new covariate representing endurance. This would help distinguish between sustainable gaits and short-term sprints.

Second, I propose collecting an additional covariate representing the amount of coaching or training the child receives (ie: hours per month). The current demographic data does not capture the variation in walking technique or the child's comfort with their gait. Since walking strategies for CP vary, a child with extensive coaching may achieve faster speeds through better technique rather than just strength alone. Controlling for this support structure would help isolate the specific impact of strength on walking speed. If possible, a more comprehensive approach would involve incorporating motion data from the gait lab sensors to classify gait technique. This would add a secondary question about which techniques are preferred for speed and endurance.

Appendix

Logistic Regression Plots

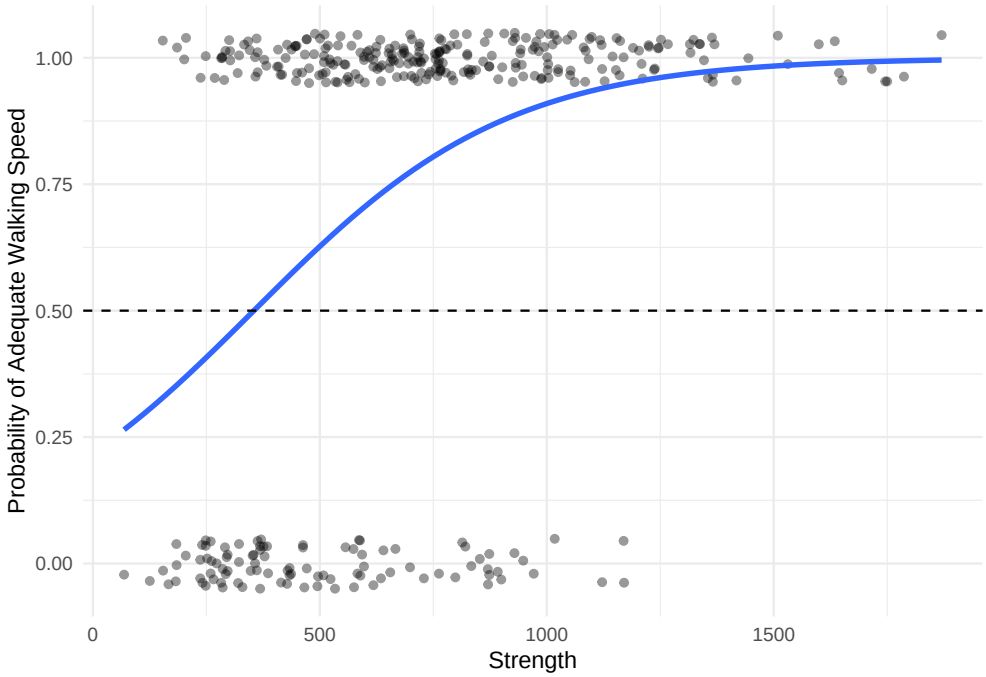

Figure 1: Probability of Adequate Walking Speed by Strength

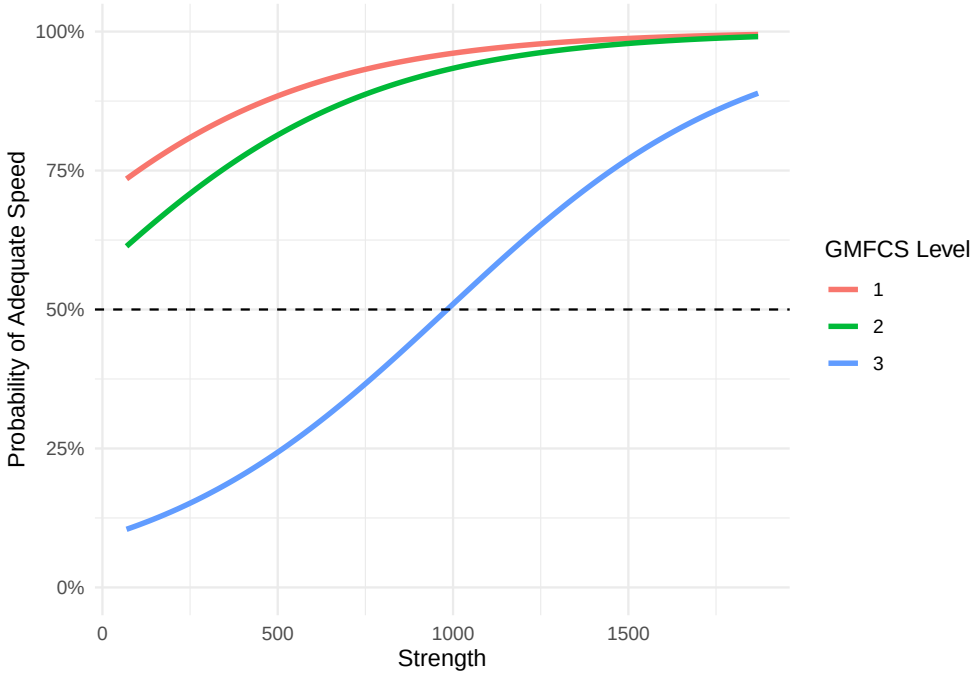

Figure 2: Predicted Probability of Adequate Walking Speed

EDA Figures

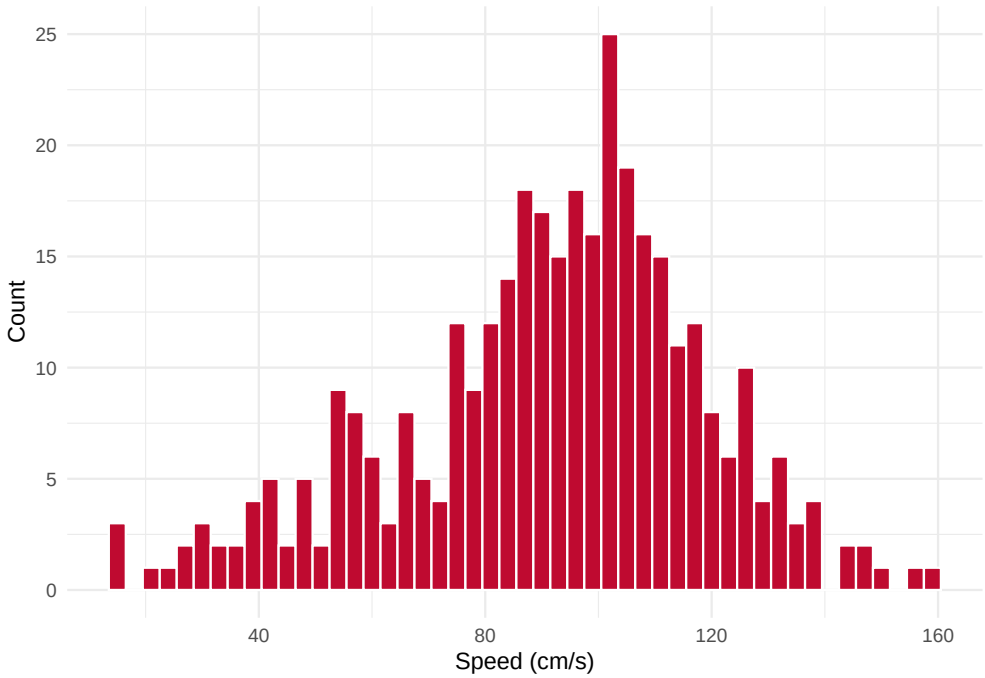

Figure 3: Distribution of Walking Speed

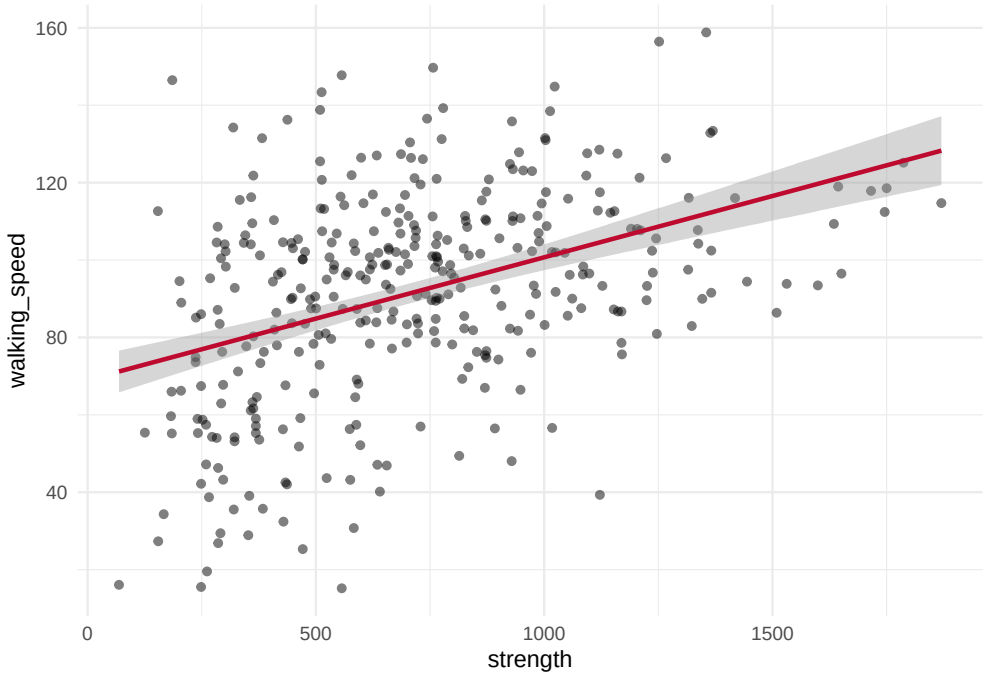

Figure 4: Strength vs Walking Speed

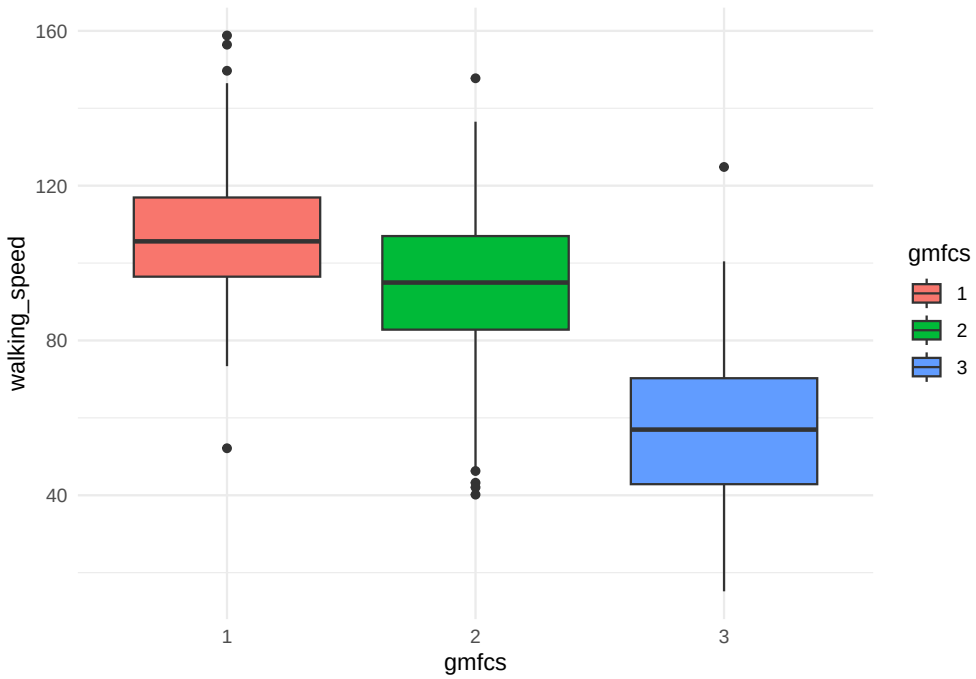

Figure 5: Walking Speed by GMFCS Level

Assumption Plots

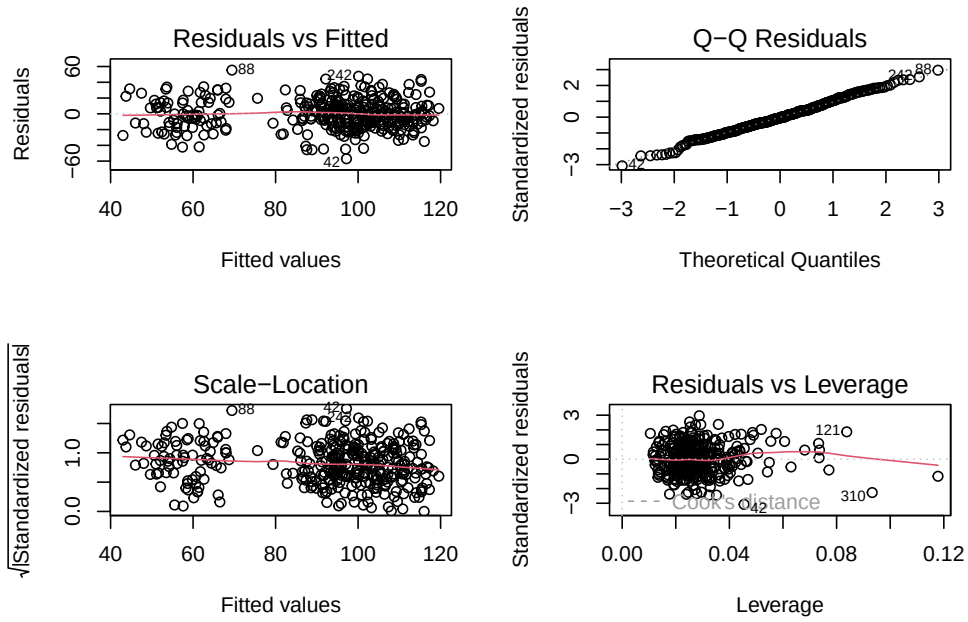

Figure 6: OLS Assumption Graphs

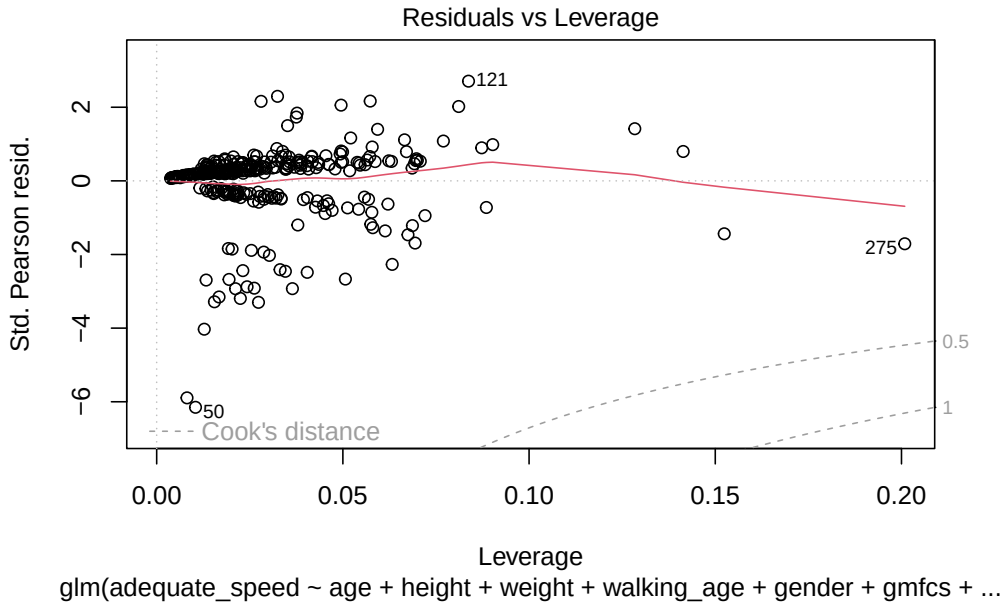

Figure 7: Logistic Regression Residuals vs Leverage

Model Summaries

Model 1

```
##
## Call:
## lm(formula = walking_speed ~ age + height + weight + walking_age +
##     gender + gmfcs + cptype, data = df_complete)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -55.487 -12.012  -0.043  12.603  60.049
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       66.17773   15.24725   4.340 1.87e-05 ***
## age               -2.80545    0.63708  -4.404 1.42e-05 ***
## height             0.55003    0.14952   3.679 0.000272 ***
## weight            -0.06126    0.09698  -0.632 0.528007
## walking_age       -0.14252    0.08197  -1.739 0.082970 .
## genderMale        -2.28436    2.17835  -1.049 0.295069
## gmfcs2            -8.92707    2.68524  -3.325 0.000981 ***
## gmfcs3           -41.79269    3.47648 -12.022  < 2e-16 ***
## cptypeHemiplegia   3.10447    2.66623   1.164 0.245084
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 19.24 on 343 degrees of freedom
## Multiple R-squared:  0.5029, Adjusted R-squared:  0.4913
## F-statistic: 43.38 on 8 and 343 DF,  p-value: < 2.2e-16
```

Model 2

```
##
## Call:
## lm(formula = walking_speed ~ age + height + weight + walking_age +
##     gender + gmfcs + cptype + strength, data = df_complete)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -57.018 -12.256  -0.742  11.896  55.418
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       67.903975  15.044790   4.513 8.79e-06 ***
## age               -2.630856   0.630489  -4.173 3.82e-05 ***
## height             0.463957   0.149768   3.098  0.00211 **
## weight            -0.105294   0.096572  -1.090  0.27634
## walking_age       -0.107418   0.081537  -1.317  0.18858
## genderMale        -2.788915   2.153619  -1.295  0.19620
## gmfcs2            -6.951607   2.715782  -2.560  0.01091 *
## gmfcs3           -38.086285   3.610171 -10.550  < 2e-16 ***
## cptypeHemiplegia   3.507489   2.632090   1.333  0.18355
## strength           0.011703   0.003573   3.275  0.00116 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.98 on 342 degrees of freedom
## Multiple R-squared:  0.518,  Adjusted R-squared:  0.5053
## F-statistic: 40.84 on 9 and 342 DF,  p-value: < 2.2e-16
```

Model 3

```
##
## Call:
## lm(formula = walking_speed ~ age + height + weight + walking_age +
##     gender + gmfcs + cptype * strength + gmfcs * strength, data = df_comple
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -56.73 -12.62   0.28  11.84  47.48
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                70.801829  15.918889   4.448 1.18e-05 ***
```

```
## age                        -2.465876   0.631498  -3.905 0.000114 ***
## height                      0.465491   0.149128   3.121 0.001955 **
## weight                     -0.134209   0.096850  -1.386 0.166735
## walking_age                -0.116346   0.081774  -1.423 0.155721
## genderMale                 -3.022393   2.143740  -1.410 0.159495
## gmfcs2                    -11.108866   6.619098  -1.678 0.094209 .
## gmfcs3                    -50.234341   7.415684  -6.774 5.55e-11 ***
## cptypeHemiplegia            6.570090   6.645152   0.989 0.323513
## strength                    0.007898   0.006478   1.219 0.223628
## cptypeHemiplegia:strength  -0.003994   0.007727  -0.517 0.605535
## gmfcs2:strength             0.004625   0.007893   0.586 0.558299
## gmfcs3:strength             0.022180   0.010466   2.119 0.034791 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.87 on 339 degrees of freedom
## Multiple R-squared:  0.5277, Adjusted R-squared:  0.5109
## F-statistic: 31.56 on 12 and 339 DF,  p-value: < 2.2e-16
```

Model 4

```
##
## Call:
## lm(formula = walking_speed ~ age + height + weight + walking_age +
##     gender + gmfcs + cptype + gmfcs * strength, data = df_complete)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -56.281 -12.557   0.294  12.097  47.790
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       73.282941  15.161736   4.833 2.04e-06 ***
## age               -2.468818   0.630791  -3.914  0.00011 ***
## height             0.461453   0.148763   3.102  0.00208 **
## weight            -0.134277   0.096745  -1.388  0.16606
## walking_age       -0.119562   0.081449  -1.468  0.14305
## genderMale        -3.036530   2.141254  -1.418  0.15708
## gmfcs2           -12.400562   6.122730  -2.025  0.04361 *
## gmfcs3           -52.017961   6.557031  -7.933 3.12e-14 ***
## cptypeHemiplegia   3.413585   2.618853   1.303  0.19330
## strength           0.005700   0.004882   1.167  0.24383
## gmfcs2:strength    0.006360   0.007135   0.891  0.37337
## gmfcs3:strength    0.024464   0.009478   2.581  0.01026 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.85 on 340 degrees of freedom
```

```
## Multiple R-squared:  0.5273, Adjusted R-squared:  0.512
## F-statistic: 34.48 on 11 and 340 DF,  p-value: < 2.2e-16
```

Model 5

```
##
## Call:
## lm(formula = walking_speed ~ age_c + height_c + weight_c + walking_age_c +
##     gender + gmfcs + cptype + gmfcs * strength_c, data = df_centered)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -56.281 -12.557   0.294  12.097  47.790
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       104.570148   2.750569  38.018  < 2e-16 ***
## age_c              -2.468818   0.630791  -3.914  0.00011 ***
## height_c            0.461453   0.148763   3.102  0.00208 **
## weight_c           -0.134277   0.096745  -1.388  0.16606
## walking_age_c      -0.119562   0.081449  -1.468  0.14305
## genderMale         -3.036530   2.141254  -1.418  0.15708
## gmfcs2             -7.864929   2.737001  -2.874  0.00431 **
## gmfcs3            -34.572308   3.977864  -8.691  < 2e-16 ***
## cptypeHemiplegia    3.413585   2.618853   1.303  0.19330
## strength_c          0.005700   0.004882   1.167  0.24383
## gmfcs2:strength_c   0.006360   0.007135   0.891  0.37337
## gmfcs3:strength_c   0.024464   0.009478   2.581  0.01026 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.85 on 340 degrees of freedom
## Multiple R-squared:  0.5273, Adjusted R-squared:  0.512
## F-statistic: 34.48 on 11 and 340 DF,  p-value: < 2.2e-16
```

Model 6

```
##
## Call:
## glm(formula = adequate_speed ~ age + height + weight + walking_age +
##     gender + gmfcs + cptype + strength, family = "binomial",
##     data = logistic_data)
##
## Coefficients:
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.9103648  2.4231147  -0.788 0.430467
## age              -0.2014679  0.1017926  -1.979 0.047794 *
## height            0.0467487  0.0242229   1.930 0.053615 .
```

```
## weight           -0.0271301  0.0151675  -1.789 0.073664 .
## walking_age      -0.0124019  0.0116433  -1.065 0.286806
## genderMale       -0.4522483  0.3673919  -1.231 0.218334
## gmfcs2           -0.5570381  0.4827941  -1.154 0.248590
## gmfcs3           -3.1676885  0.5586771  -5.670 1.43e-08 ***
## cptypeHemiplegia  0.5941381  0.5107619   1.163 0.244733
## strength          0.0023483  0.0006738   3.485 0.000492 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 414.46  on 351  degrees of freedom
## Residual deviance: 243.18  on 342  degrees of freedom
## AIC: 263.18
##
## Number of Fisher Scoring iterations: 6
```

Model 7

```
##
## Call:
## glm(formula = adequate_speed ~ age + height + weight + walking_age +
##     gender + cptype + strength * gmfcs, family = "binomial",
##     data = logistic_data)
##
## Coefficients:
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.6836119  2.6219413  -0.642 0.520793
## age              -0.2160598  0.1027057  -2.104 0.035407 *
## height            0.0494185  0.0245147   2.016 0.043814 *
## weight           -0.0294778  0.0147500  -1.998 0.045663 *
## walking_age      -0.0162077  0.0117576  -1.378 0.168052
## genderMale       -0.5220053  0.3805488  -1.372 0.170152
## cptypeHemiplegia  0.5322228  0.5084097   1.047 0.295174
## strength          0.0021216  0.0013786   1.539 0.123823
## gmfcs2           -0.0835168  1.2063506  -0.069 0.944806
## gmfcs3           -4.6027795  1.3752863  -3.347 0.000818 ***
## strength:gmfcs2  -0.0008014  0.0015841  -0.506 0.612892
## strength:gmfcs3   0.0023750  0.0018424   1.289 0.197362
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 414.46  on 351  degrees of freedom
## Residual deviance: 238.31  on 340  degrees of freedom
## AIC: 262.31
```

```
##
## Number of Fisher Scoring iterations: 6
```

References

Kim, Bommae. 2016. *Hierarchical Linear Regression.* University of Virginia.

Kulak, Wojciech and Wojciech Sobaniec. 2004. *Comparisons of right and left hemiparetic cerebral palsy.* Pediatric Neurology.